

Shannon's Source Coding Protocol

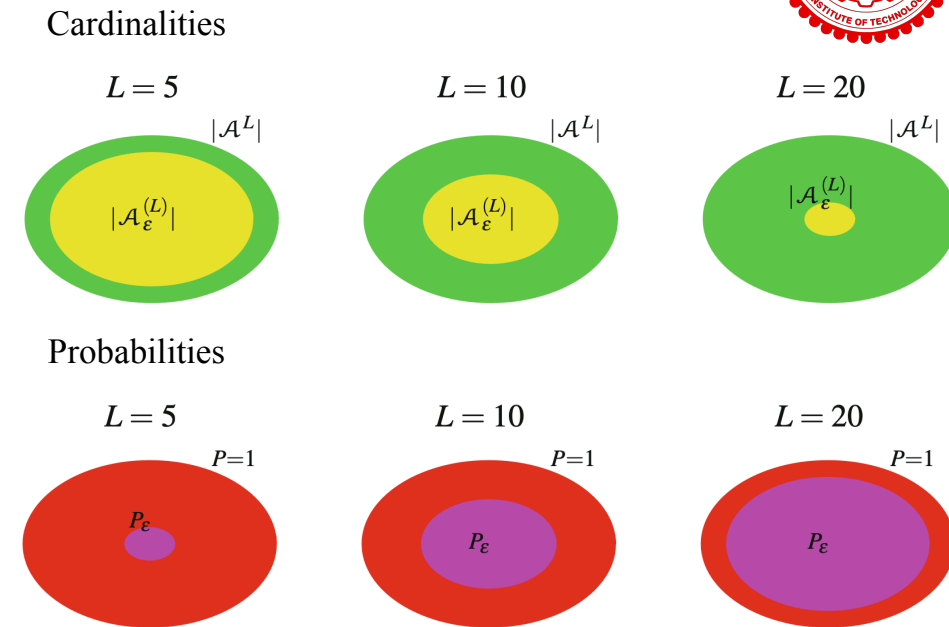


- **Asymptotic equipartition property (AEP):** Set of the possible sequences emitted by source is subdivided into two distinct classes, called *typical sequences* and *atypical sequences*.
- Example: IID binary source $P[A = 1] = p$ and $P[A = 0] = q = 1 - p$. For an L length sequence we expect to see in the sequence approximately Lp 1s, and Lq 0s (Typical Sequence)
- Probability of Typical sequence: $p(a^L) = p_A(a_1) \cdots p_A(a_L) \simeq p^{Lp} q^{Lq}$ and information: $i(a^L) \simeq -Lp \log p - Lq \log q = LH(A)$. Thus, $p(a^L) \simeq 2^{-LH(A)}$
- **Typical Sequences:** (1) are equally likely (uniformly distributed), and (2) their number is at most $2^{LH(A)}$
- **Shannon's Data Compression:** There are at most $N \simeq 2^{LH(A)}$ typical sequences, it is sufficient to create a codebook, or an index set, of cardinality N , where each codeword can be represented by a string of $\log N \simeq LH(A)$ bits, i.e., using $H(A)$ bits/symbol.
- **Definition:** A realization a^L is ε -typical if its sample entropy $\mathcal{J}(a^L) = -\frac{1}{L} \log p_A(a_1) \cdots p_A(a_L) = -\frac{1}{L} \sum_{n=1}^L \log p_A(a_n)$ is ε -close to the entropy $H(A)$, i.e., $H(A) - \varepsilon \leq \mathcal{J}(a^L) \leq H(A) + \varepsilon$.
- The ε -typical set $A_\varepsilon^{(L)}$ is the set of all ε -typical sequences s.t. $2^{-L(H(A)+\varepsilon)} \leq p(a_1) \cdots p(a_L) \leq 2^{-L(H(A)-\varepsilon)}$



Shannon's Compression Protocol

- Pictorially: ϵ -typical sequences.
- A compression or coding protocol $\mathcal{C}_L$ maps the sourceword $A^L = (A_1, \dots, A_L) \in \mathcal{A}^L$ into binary codeword $C^{LR} = (C_1, \dots, C_{LR})$ of LR bits.
- A decompression or decoding protocol $\mathcal{D}_L$ maps the LR bits back into a sequence $\tilde{A}^L = \mathcal{D}(C^{LR})$ of the original alphabet.
- Coding map: $\mathcal{C}_L : \mathcal{A}^L \rightarrow \{0,1\}^{LR}$ and decoding map: $\mathcal{D}_L : \{0,1\}^{LR} \rightarrow \mathcal{A}^L$
- Reliable protocol if: $\mathcal{D}_L(\mathcal{C}_L(A^L)) = A^L$ (*Asymptotically reliable* if this holds with probability one as $L \rightarrow \infty$)



The fraction of ϵ -typical sequences decreases with L .
 The probability P_ϵ of ϵ -typical sequences increases with L

- The parameter R is called the *compression rate* and represents the number of bits/symbol used in the encoding.
- **Shannon's Noiseless Channel Coding Theorem:** Consider an IID source with entropy $H(A)$ and a compression rate of R bit/symbol. Then, if $R > H(A)$, there exists an asymptotically reliable compression protocol. Conversely, if $R < H(A)$ any compression protocol is not reliable.
- **Proof:** Application of properties of **Typical Sequences**



Shannon's AEP v/s Real Compression Algorithms

- Shannon's typical sequence (asymptotic equipartition property, AEP) is a theoretical foundation for compression, but it doesn't translate into a practical coder-decoder (codec).
- Requirements of “Typical-sequence compression”:
 - For a length- L source, identify the typical set (about 2^{LH} sequences)
 - Build a one-to-one mapping between each typical sequence and an index of $\approx LH$ bits.
 - Encoding: recognize which exact length- L sequence you have & output its index; Decoding: invert the map.
- Drawbacks of Shannon's AEP approach:
 - Need to enumerate/store an exponential number of sequences.
 - Need to know the true source distribution $P(x)$
 - Operate on very large blocks
 - Not practical: Requires exponential complexity and storage
- Practical Codecs (JPEG, PDF, etc) need low latency and streaming operation (byte-by-byte, block-by-block)
 - Uses the spatial redundancy, transforms and sparsity properties of Images and documents.
 - Practical codecs learn the probabilities online and handle non-stationarity of data

AEP gives the limit H (what's achievable)

Practical codecs give algorithms $\approx H$ (how to achieve it efficiently)

Overview

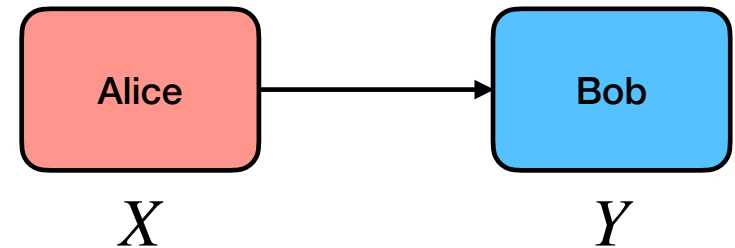


- Introduction
- Classical Entropy
- Noiseless Source Coding: Data Compression
- **Noisy Channel Coding: Channel Capacity**

Conditional Entropy: Motivation



- Let X and Y be two random variables.
- $H(X)$ measures our uncertainty about X .
- If we observe Y , our uncertainty about X can reduce.
- This reduced uncertainty is quantified as the **Conditional Entropy: $H(X|Y)$** .
- Example Scenario:
 - Alice sends X to Bob over a noisy channel.
 - Bob receives Y , a noisy version of X .
 - Bob wants to estimate X given Y .





Conditional Entropy: Definition

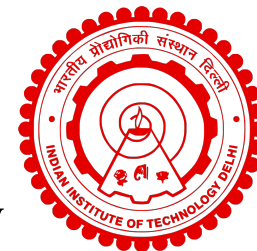
- **Given:** Joint distribution $p_{X,Y}(x, y)$ then $H(X|Y = y) = - \sum_x p_{X|Y}(x|y) \log p_{X|Y}(x|y)$

- Then, the **conditional entropy** is:

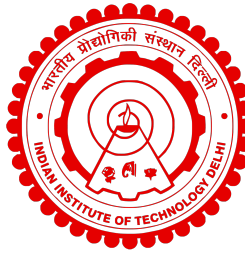
$$\begin{aligned} H(X|Y) &= \sum_y p_Y(y) H(X|Y = y) \\ &= - \sum_y p_Y(y) \sum_x p_{X|Y}(x|y) \log p_{X|Y}(x|y) \\ &= - \sum_{x,y} p_{X,Y}(x, y) \log p_{X|Y}(x|y) \end{aligned}$$

- **Special Case:** If $X \perp Y$, then $H(X|Y) = H(X)$

Conditioning Does Not Increase Entropy



- **Theorem:** For any two random variables X and Y , $H(X) \geq H(X|Y)$ with equality if and only if X and Y are independent.
- **Intuition:**
 - $H(X)$ represents the uncertainty about X before observing Y .
 - $H(X|Y)$ represents the remaining uncertainty after observing Y .
 - Gaining information (even noisy) about X via Y should not increase uncertainty.
- **Equality Condition:** $H(X) = H(X|Y)$ if and only if X and Y are independent, i.e., $p_{X,Y}(x, y) = p_X(x)p_Y(y)$
- **Interpretation:** Conditioning reduces or preserves entropy. It never increases it.



Joint Entropy

- Let X and Y be two random variables with joint distribution $p_{X,Y}(x, y)$.

- The **Joint Entropy** is: $H(X, Y) = - \sum_{x,y} p_{X,Y}(x, y) \log(p_{X,Y}(x, y))$

- **Chain Rule:** $H(X, Y) = H(X) + H(Y|X)$

- **Proof:** Using the identity $p_{X,Y}(x, y) = p_{Y|X}(y|x)p_X(x)$, we get:

$$\begin{aligned} H(X, Y) &= - \sum_{x,y} p_{X,Y}(x, y) \log \left(p_{Y|X}(y|x)p_X(x) \right) \\ &= - \sum_{x,y} p_{X,Y}(x, y) \log(p_X(x)) - \sum_{x,y} p_{X,Y}(x, y) \log(p_{Y|X}(y|x)) \\ &= H(X) + H(Y|X) \end{aligned}$$

- By symmetry: $H(X, Y) = H(Y) + H(X|Y)$
- Chain Rule shows how knowing one variable reduces uncertainty about the pair.



Mutual Information

- Let X and Y be discrete random variables with joint distribution $p_{X,Y}(x, y)$
- **Mutual Information** is defined as:

$$I(X : Y) = H(X) - H(X|Y) = H(Y) - H(Y|X)$$

- **Interpretation:**
 - $H(X)$: Uncertainty of X .
 - $H(X|Y)$: Remaining uncertainty of X given Y .
 - $I(X : Y)$: Amount of information X and Y share.
- **Properties:**
 - If X and Y are independent, then: $H(X|Y) = H(X) \Rightarrow I(X : Y) = 0$
 - **Theorem:** Mutual information is always non-negative: $I(X : Y) \geq 0$ with equality iff X and Y are independent.



Conditional Mutual Information

- Given a third variable Z , the **Conditional Mutual Information** is:

$$I(X : Y|Z) = H(Y|Z) - H(Y|X, Z)$$

- Alternative form: $I(X : Y|Z) = H(X|Z) - H(X|Y, Z) = H(X|Z) + H(Y|Z) - H(X, Y|Z)$
- Chain Rule for Mutual Information: Let $X_1, X_2, \dots, X_n$ and Y be random variables. Then:

$$I(X_1, \dots, X_n : Y) = \sum_{i=1}^n I(X_i : Y|X_1, \dots, X_{i-1})$$

- This expresses mutual information as an accumulated quantity.